

		Energy transferred i.e. work done on charge = $QV = 5.0 \times 2700 = 13500 \text{ J}$
22	C	Initially, when distance x (less than mid-way of OZ from O) increases, with a small change in distance x , the cross-sectional area increases, resulting in smaller increases in R . Later, when distance x (more than mid-way of OZ from O) increases, with a small change in distance x , the cross-sectional area decreases, resulting in larger increases in R .
23	C	Since potentials at points C and B are equal, we can remove the resistor between them